Analysis Plan — Peripheral High-Frequency Activity and the Scalp Aperiodic Exponent (HBN Release 3)

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Status and purpose

This plan was fixed on 20 October 2026, before any cohort-level outcome was inspected. Only three pilot children (NDARAA948VFH, NDARAC853DTE, NDARAD774HAZ) have been processed; they shaped the quality-control rules but were used for no hypothesis test.

All features come from pipeline v3.1 (`pilot_v31.py`, fooof 1.1.1), run on HBN Release 3 with `run_r3_streaming.py`. Any later change goes in the deviations log, with its date and reason, before dependent results are inspected.

Research questions and hypotheses

The study tests whether peripheral high-frequency activity is associated with the scalp aperiodic exponent — but only after showing that the peripheral measure tracks muscle.

1. **Q1 — gatekeeper.** Is the peripheral index a reliable, spatially specific high-frequency signal — the signature expected of muscle? Its result decides which branch runs.
2. **Q2 — primary (if Q1 passes).** Is peripheral high-frequency activity (PHF) associated with the scalp exponent? Expected direction: negative, because muscle flattens spectra; tested two-sided.
3. **Q3 — secondary (if Q1 passes).** Does adjusting for PHF change the association between the exponent and the CBCL attention factor? Estimated, not tested.
4. **Fallback (if Q1 fails).** How much does channel-level high-frequency contamination shift the exponent, and does standard fit-quality filtering hide it?

The CBCL attention factor is a symptom dimension, not an ADHD diagnosis.

Data and sample

The sample is every HBN Release 3 child (OpenNeuro ds005507; 184 listed) whose resting-state recording the pipeline processes successfully.

- **Included:** curator RestingState status `available` (142) and `caution` (42). Status is recorded for a sensitivity check.
- **Excluded from all analyses:** processing failure; fewer than 80 reference-eligible scalp channels; under 120 s of eyes-closed data. The pilot children had 100–108 channels and 180 s.
- **Excluded from Q3 only:** children missing the attention factor (4 of 184 in `participants.tsv`).

With 150–180 children, Q2 has 80% power to detect r ≈ 0.21–0.23 (two-sided α = 0.05).

Measures

All measures are per child, from eyes-closed data, with channels classified by the frozen v3.1 quality-control rules.

Role	Variable	Definition
Outcome	Scalp exponent	`scalp__ap2_exponent`: median fixed-mode exponent, 2–40 Hz, reference-eligible scalp channels with $R^2 \geq 0.85$
Predictor	PHF	Mean of log10(`face_jaw__rel_70_110`) and log10(`neck__rel_70_110`). Infraorbital channels excluded (eye contamination)
Covariate	Age	Natural cubic spline, 3 df
Covariate	Sex	As recorded
Q3 exposure	Attention	CBCL attention factor (`attention`), continuous

The exponent is treated as a spectral parameter; an excitation–inhibition interpretation is not assumed.

Gatekeeper (Q1)

The peripheral index counts as a reliable, spatially specific high-frequency signal only if both criteria hold; otherwise the fallback runs.

1. **Reliable.** Split-half reliability of PHF ⩾ 0.70: PHF from eyes-closed blocks 1, 3, 5 against blocks 2, 4, as a Pearson correlation across children, Spearman–Brown corrected.
2. **Spatially specific.** Spearman correlation between PHF and the within-child difference log10(`frontal__rel_70_110`) − log10(`scalp_nonfrontal__rel_70_110`); its 95% bootstrap CI must lie entirely above 0.

A noise floor shared across the scalp cancels in the within-child difference, so it cannot satisfy criterion 2. A floor stronger over frontal than posterior channels can; these data cannot exclude that, so passing establishes spatial specificity, not muscle (deviation D1). Frontal values include high-variance channels by design.

Primary analysis (Q2)

One test carries the primary claim: the PHF coefficient in a robust regression, run only if Q1 passes.

```
\text{exponent} = \beta_0 + \beta_{\text{PHF}}\,\text{PHF} + s(\text{age}) +
\beta_{\text{sex}}\,\text{sex} + \varepsilon
```

- Huber M-estimator; exponent and PHF standardised, so β is in standard-deviation units.
- 95% CI from 2,000 bootstrap resamples of children (percentile method); significance at two-sided α = 0.05.
- Also reported: Spearman ρ(PHF, exponent), and the partial ρ adjusting for age and sex.

Secondary analysis (Q3)

Q3 estimates how far the attention association moves when PHF is added; it makes no significance claim.

- **Model A:** exponent ~ attention + s(age) + sex. **Model B:** Model A + PHF.
- Report β_attention from both, and Δβ = β_A − β_B with a 2,000-resample bootstrap CI.
- Complete cases only.

A small β_attention is expected: the 2026 HBN registered report found no reliable ADHD difference in the exponent. Whatever is found is reported as found.

Fallback analysis (if Q1 fails)

If the gatekeeper fails, the question becomes how much high-frequency contamination shifts the exponent, and whether standard filtering hides it.

1. **Primary:** per child, D = `scalp_allvalid__ap2_exponent_nofilter_mean` − `scalp__ap2_exponent`. Test median D ≠ 0 with a Wilcoxon signed-rank test, plus a 2,000-resample bootstrap CI.
2. **Secondary:** Spearman ρ between D and log10(`frontal__rel_70_110`).
3. **Secondary:** β_attention estimated with the standard and with the uncensored exponent; report the difference with a bootstrap CI.

Fit-failure and high-variance channel counts per child are reported descriptively.

Sensitivity analyses

Each re-runs the analysis of whichever branch ran. Results are descriptive and cannot create new claims; an item that duplicates the branch's own variables is skipped.

1. Excluding `caution` recordings.
2. 1–40 Hz fixed-mode exponent (`scalp__ap1_exponent`).
3. Knee-mode exponent (`scalp__ap1k_exponent`).
4. Uncensored exponent (`scalp_allvalid__ap2_exponent_nofilter_mean`).
5. PHF from neck channels only, then face/jaw only.
6. 30–45 Hz relative power in place of 70–110 Hz.
7. Adding the scalp noise-floor descriptor (`scalp__hf_excess_over_floor`) as a covariate.

Statistical conventions

- **Unit of analysis:** the child. Windows, blocks and channels are never treated as independent observations.
- **Random seed:** 20261020 for all bootstrap resampling.
- **Missing data:** complete cases per analysis; counts reported at each step.

- **Outliers:** handled by the Huber estimator; no removal beyond the stated exclusions.
- **Multiplicity:** one primary test (Q2, or fallback item 1). Q3, fallback secondaries and sensitivity analyses are estimation only.
- **Software:** pipeline versions are recorded in `R3_v31_provenance.json`; the analysis script is versioned with it.

Interpretation guardrails

- Associations are never described as causal.
- Even if Q1 passes, muscle is the leading interpretation, not an established fact: an uneven noise floor can also satisfy criterion 2. The signal is never called EMG.
- The exponent is a spectral parameter, not a measure of excitation–inhibition balance.
- Attention results describe a CBCL symptom dimension, not ADHD diagnosis.
- Units are taken as μV from BIDS metadata; the acquisition-to-export conversion is unverified, so absolute-amplitude measures stay secondary.

Deviations log

Two deviations are recorded. D1 was made before any cohort data was inspected; D2 corrects a coding error found after the analysis ran. Later changes are entered here before dependent results are inspected.

Date	Change	Reason
2026-10-20	D1: criterion 2 changed from a partial Spearman correlation controlling non-frontal high-frequency power to a correlation with the frontal-minus-non-frontal difference. The muscle claim was narrowed to spatial specificity.	In simulation, the original criterion passed a muscle-free cohort ($\rho = 0.59$) through residual confounding from a noisy floor proxy; the new one rejected it ($\rho = -0.11$). Neither excludes a floor stronger over frontal channels.

Date	Change	Reason
2026-10-21	D2: in fallback secondary analysis 3, the attention coefficient is now selected by name. The script had selected design column 1, which is the sex term.	Coding error: the patsy design places categorical terms first, so the previously reported values (−0.054, −0.116, difference +0.062) were sex coefficients. The planned analysis is unchanged; only its implementation is corrected. All other results are unaffected.